

Assignment 2: Dangerousity

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Introduction

Match statistics in Australian Football record discrete events. An inside 50 kick is counted identically, whether it produces a mark 20 metres directly in front of goal, or a hurried kick into a three-on-one contest. Player tracking data makes it possible to assess how dangerous a position is at every moment of a passage of play, not just the moments that end in a shot at goal.

Link et al. (2016) proposed Dangerousity as a real-time measure of threat in association Football. They defined it as the probability of the current situation resulting in a goal. The measure combines four terms describing the location of the ball, the ball carrier's control of it, the pressure applied by nearby defenders, and the density of surrounding players. Location sets a ceiling of the probability of a goal, and the other three terms can only reduce it. Because the dangerousity value updates continuously, Link et al. (2016) were also able to isolate and quantify the value of individual actions within a given passage of play.

This report proposes a conceptually equivalent measure that is specific for Australian Football, rather than a direct translation of Link et al.'s (2016) measure. Australian Football differs from association Football in ways that change what makes a position dangerous. There is no crossbar, so a shot cannot be too high. A defender may legally tackle from any direction, including from behind (AFL, 2022). A set shot suspends defensive interference entirely, so the shooter chooses distance, angle and technique unopposed. Each difference is addressed in the sections that follow.

Dangerousity (DA) is defined here as the probability that the attacking team kicks a goal from the situation shown in a single frame of tracking data. The available data provided one frozen frame per play and no subsequent events. DA therefore describes the value of the position, not the probability that a particular passage of play ends in a goal.

The formula was constructed through reasoning from the rules and structure of the game, rather than by fitting parameters to the outcomes. The primary source is Anderson et al. (2018), whose analysis of 4,599 set shots from a full AFL season provided the conversion rates used here. The model estimates the probability of a goal only. Behinds, match context and player identity fall outside its scope and are discussed as extensions later in the report.

The Formula

Dangerousity is calculated from a single frame of tracking data as:

$$DA = LO \times \left(1 - \frac{PR + (1 - SI) \cdot ST}{k_1} \right) \quad \text{Equation 1.}$$

where LO is Location, PR is Pressure, ST is Structure of defenders and forwards, and SI is Shot Intent. All four terms are bounded between 0 and 1. The constant k_1 bounds the size of the discount. Figure 1 shows how each term is derived from the frame and how they combine.

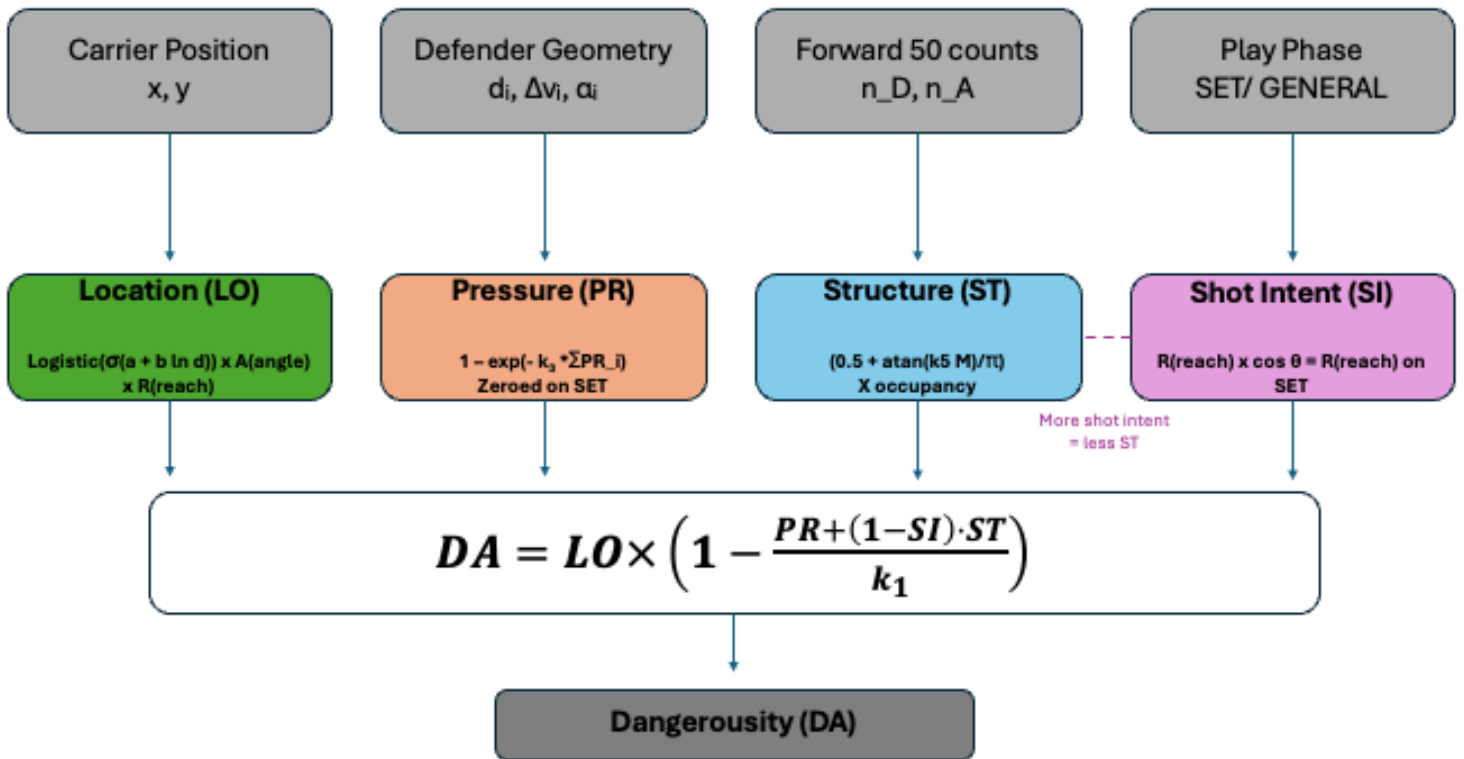


Figure 1. Structure of the Dangerousity model. The top row lists the quantities taken from a single frame. The middle row shows the four terms derived from them.

Why this form

The terms are combined multiplicatively rather than additively. Location is the only term that represents the probability of scoring a goal. Pressure and Structure capture how much of that probability the opposition removes. An additive model would allow the defensive terms to generate danger even where location contributes none, meaning a heavily pressured player 90 meters from goal could score higher than an unpressured player just 20 metres out. The multiplicative form prevents this. Location sets a ceiling on the danger value, and the remaining terms can only lower it.

The constant k_1 bounds how far Pressure and Structure can lower that value. Link et al. (2016) used the same device, choosing their constant so that location could be reduced by no more than one half. The bracketed term in Equation 1 has a maximum value of 2, so $k_1 = 2.5$ permits a reduction to one-fifth of the location value.

Constants are numbered to match Link et al. (2016), so that k_1 , k_3 and k_5 denote the same quantities in both models. No k_2 or k_4 appears here. Control has no counterpart in this model. Density is only partly retained: Link et al. (2016) derived it from the probability of a shot being blocked and the probability of a pass being intercepted, but only the interception component applies here.

Model Inputs

The model requires the position of every attacking and defending player, the identity of the player in possession, the speed of each player, and an indicator of whether the play is a set play (mark/free kick) or general play. It uses no other information, so it can be applied to any frame in which these quantities are recorded.

The dataset used here contains 102 plays. Each play is a single frame recording 36 players, 18 attacking and 18 defending, with exactly one player identified as being in possession. Positions are given in metres, and speeds in metres per second. Sixty plays are set plays, and forty-two are general play. Each play carries one timestamp, so the data records nothing about what preceded the frame or what followed it.

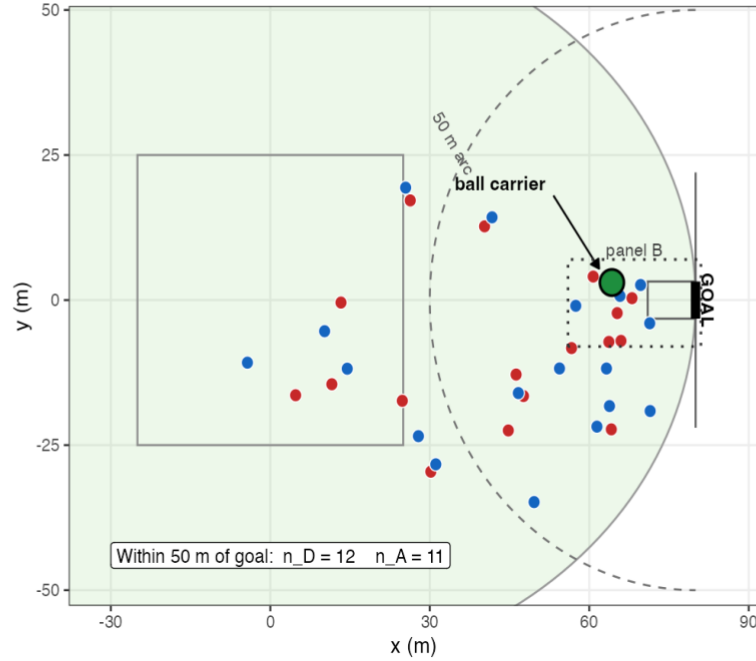
Positions are recorded on a ground measuring 160 m by 129 m, centred on the origin. The attacking team moves in the direction of increasing x , and all plays are oriented this way. The goal under attack is therefore centred at $(80, 0)$, with the goal posts at $y = 3.2$ and $y = -3.2$.

The ball carrier lies between 1.6 m and 91.2 m from goal, with a median of 59.1 m. Only 27 of the 102 carriers are within 50 m. Forty-two of the 60 set plays occur beyond 50 m, where a mark or free kick is kicked to a teammate rather than a shot at goal. Most frames therefore capture play from which a shot may eventually follow, rather than a shot itself. This is the reason Dangerosity is defined as the value of a position.

Three quantities relevant to how dangerous a play is are not recorded. Speed is recorded only as a magnitude with no direction, so the rate at which a defender is closing on the ball carrier cannot be calculated. Players carry an identifier, but no descriptive attributes, so player quality cannot be represented. The score and the time remaining in the game are not given.

A. One frame: all 36 players

Attackers red, defenders blue, ball carrier green. Attack is towards increasing x.



B. Geometry taken from the frame

Carrier terms in black, defender terms in blue. One defender is annotated; all 18 are used.

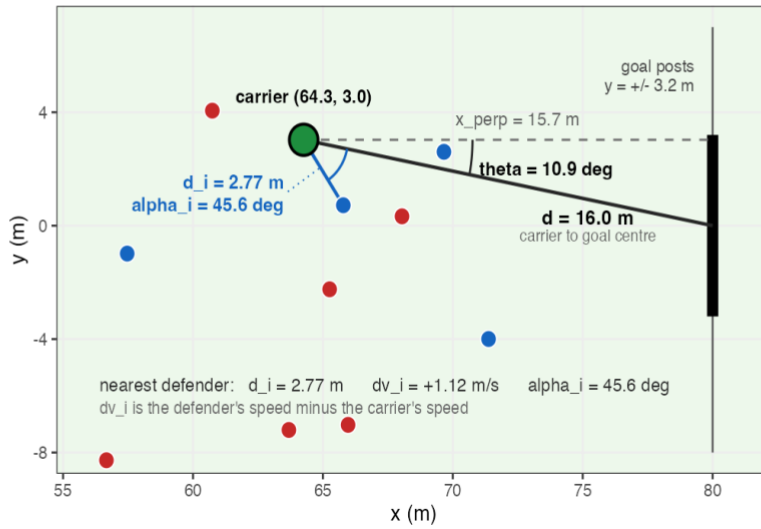


Figure 2. Model inputs, illustrated on play 9128. (A) A single frame containing all 36 players. The dashed arc marks 50 m from goal, within which n_D defenders and n_A attackers are counted. (B) The geometry derived from the frame. From the ball carrier, $x_{\perp}$ is the perpendicular distance to the goal line, d is the distance to the centre of goal, and ϑ is the angle off centre. For each defender, d_i is the distance to the carrier, α_i is the angle between the direction to that defender and the direction to goal, and Δv_i is the defender's speed minus the carrier's. One defender is annotated for clarity; the model uses all 18.

Table 1. Quantities derived from a single frame.

Symbol	Definition	Source
x, y	Position of the ball carrier	Data
$x_{\perp}$	$80 - x$, perpendicular distance to the goal line	Derived
d	$\sqrt{(x_{\perp}^2 + y^2)}$, distance to the centre of goal	Derived
θ	$\arctan(y / x_{\perp})$, angle off centre; $\theta = 0$ is directly in front	Derived
d_i	Distance from defender i to the ball carrier	Derived
Δv_i	$v_i - v$, defender speed minus carrier speed	Derived
α_i	Angle between the carrier-to-defender and carrier-to-goal directions; $\alpha_i = 0$ is directly in front of the carrier, 180° directly behind	Derived
n_D, n_A	Defenders and attackers within 50 m of the centre of goal	Derived
M	$n_D - n_A$	Derived
phase	Mark or free kick, or general play	Data

Constants

The model contains 15 constants, listed in Table 2. Five are calibrated against published set shot conversion rates. Three are taken from the source model or from the laws of the game. Three are set from the distributions observed in the data. The remaining four rest on reasoning alone.

Table 2. Constants used in the model.

Symbol	Value	Role	Basis
a	7.181	Intercept of the distance term	Fitted to set shot conversion by distance (Anderson et al., 2018)
b	-1.923	Slope of the distance term on log d	Fitted to set shot conversion by distance (Anderson et al., 2018)
p	0.7	Exponent of the angle penalty	Calibrated to the angle split reported by Anderson et al. (2018)
d ₅₀	66 m	Distance at which reachability falls to one half	Set so Location at 54 m matches the 35.5% conversion beyond 50 m (Anderson et al., 2018)
s	5	Width of the reachability transition	Set with d ₅₀ against the same conversion rates
r _H	6.0 m	Pressure radius for a defender in front of the carrier	Reasoned disposal window of 1.5 s times the observed median defender speed of 4.06 m/s
r _L	4.5 m	Pressure radius for a defender beside the carrier	Proportional to r _H , following the zone ratios of Link et al. (2016)
r _B	4.0 m	Pressure radius for a defender behind the carrier	Raised above Link's proportional value because a tackle from behind is legal (AFL, 2026)
r ₀	1.0 m	Radius within which pressure is maximal	High pressure zone of Link et al. (2016)
λ	1.0	Extent to which a closing defender stretches his radius	Reasoned
v ₀	3.0 m/s	Reference speed difference for that stretch	Set from the observed range of Δv _i
k ₁	2.5	Bounds the total discount applied to Location	Reasoned; bounds Location at one fifth of its value
k ₃	1.0	Rate at which pressure saturates as defenders accumulate	Reasoned
k ₅	0.2	Steepness of the response to a numbers advantage	Set from the observed range of M
n ₀	3	Scale for forward 50 occupancy	Reasoned

Location (LO)

Location is calibrated against set shot conversion, which is measured conditionally on a shot being taken. It therefore approximates the value of a position by the quality of the shot it would produce. Positions from which the ball is more likely to be moved to a better placed teammate before a shot is taken are understated by this approximation.

Link et al. (2016) represented location as Zone, a lookup table over a grid of 2 m squares starting 34 m from the goal line. Values in that grid were derived from five qualitative assumptions about

where a goal is likely, and no underlying equation was published. Australian Football permits a different approach because conversion rates have been measured directly across a full season of professional matches. Location is built from three components: a distance term calibrated to these measurements, an angle penalty, and a reachability term reflecting whether a shot is available at all.

Distance (D)

Anderson et al. (2018) recorded outcomes for 4,599 set shots across the 2012 AFL season, reporting conversion rates by distance band, separately for key forwards and all other players. The rates used here pool both groups, weighted by the number of shots in each, and decline from 96.2% inside 15 m to 35.5% beyond 50 m. They fitted no functional form to those rates, so the curve used here is derived from the published bands.

A straight line cannot describe these rates, because probability is bounded while a line is not. Fitted to the five bands, a linear model exceeds certainty below 3.8 m and produces a negative probability beyond 81.9 m. This second failure lies within the range of this dataset, where the ball carrier is as far as 91.2 m from goal.

The relationship is therefore fitted in log-odds, which is unbounded, so no combination of parameters can return an impossible value. Distance enters as its logarithm, since moving from 40 m to 50 m alters the probability of a goal far more than moving from 70m to 80m. Under the fitted curve, the two ten-metre changes cost 10.6 and 4.8 percentage points, respectively. This gives

$$\sigma(a + b \ln d), \quad \sigma(z) = \frac{1}{1 + e^{-z}} \quad \text{Equation 2.}$$

Where σ is the logistic function, which converts log-odds back to a probability. Weighting each band by its number of shots gives $a = 7.181$ and $b = -1.923$. The fitted curve reproduces every band to within 1.7 percentage points (Figure 3).

A straight line leaves the range a probability can occupy

Distance term only. A reachability term, introduced next, removes the long-range tail.

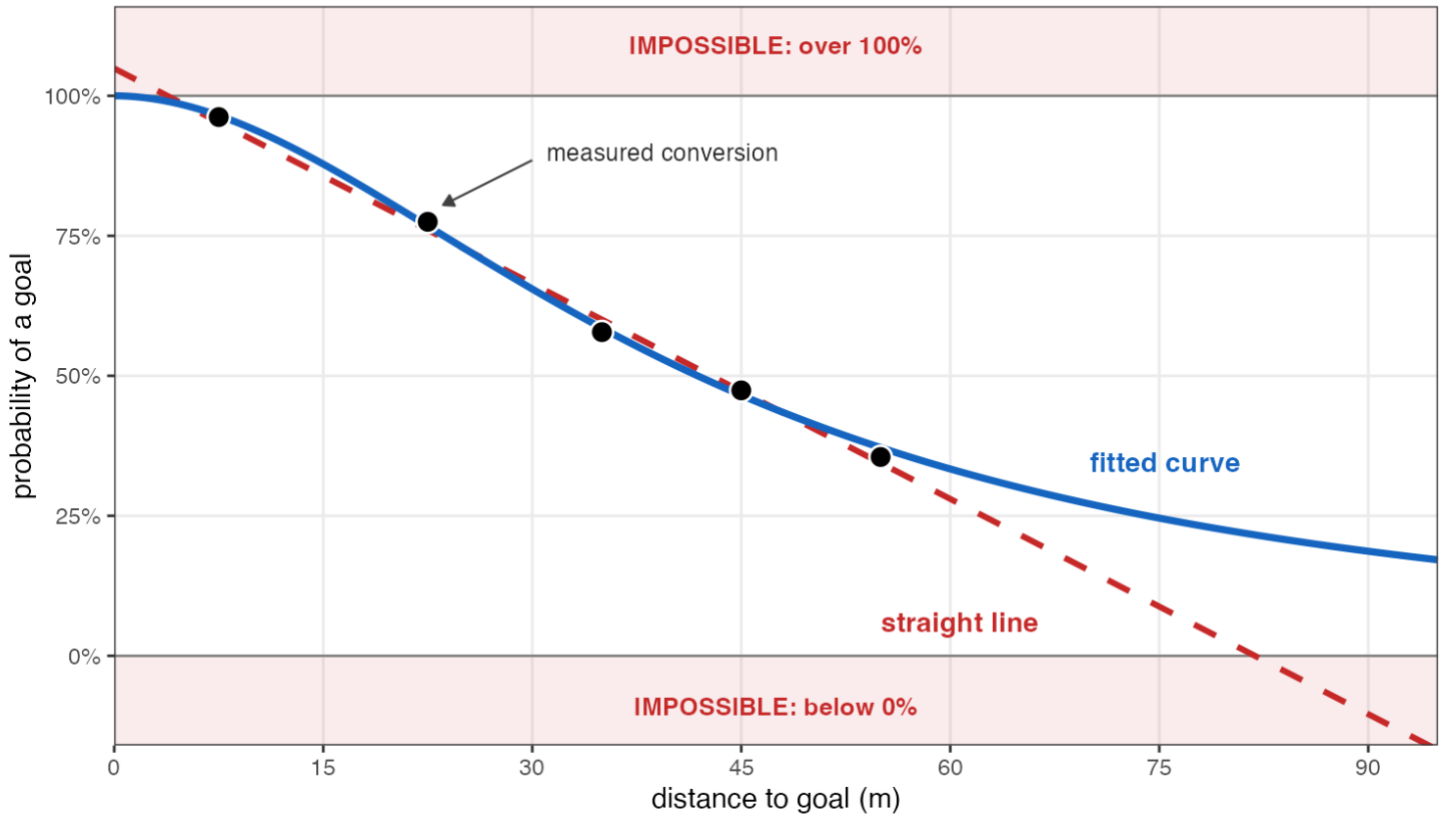


Figure 3. Set shot conversion by distance. Points are the rates reported by Anderson et al. (2018), pooled across player type, and the blue curve is the fitted model $\sigma(7.181 - 1.923 \ln d)$, which matches every band to within 1.7 percentage points. A weighted linear fit to the same rates exceeds a probability of 1 below 3.8 m and falls below 0 beyond 81.9 m.

Two limitations follow from the source. First, Anderson et al. (2018) measured set shots only, so the curve describes a player with time to prepare. Second, their final band is open above 50 m, so the curve is extrapolated beyond that distance for 75 of the 102 plays.

Angle (A)

A shot from directly in front faces the full width of the goal, while a shot from wide faces a narrower opening because the posts appear closer together as the player moves off centre.

The goal subtends an angle ω_P at the ball carrier, measured between the two posts. This is compared with ω_0 , the angle subtended from directly in front at the same perpendicular distance from the goal line. The penalty is the ratio of the two, raised to an exponent:

$$A = \left[\min \left(1, \frac{\omega_P}{\omega_0} \right) \right]^p \quad \text{Equation 3.}$$

Comparing against a reference at the same perpendicular distance is deliberate. The angle that the goal subtends narrows with range as well as with displacement, and range is already accounted for by Equation 2. Taking the ratio isolates the penalty for being off-centre, so the two terms don't both charge for the same effect. The minimum function acts as a safeguard: since the subtended angle is greatest directly in front at any given perpendicular distance, the ratio cannot exceed one.

Anderson et al. (2018) reported 63.1% accuracy for shots less than 30 degrees from goal and 39.6% at 30 degrees or more, a ratio of 0.628. Reproducing that ratio fixes p only when a representative angle is assumed for each group, and the split does not supply one. The acute

group is open above 30 degrees, so it cannot be represented by its own boundary. Taking the angle as typically 45 degrees gives $p = 0.7$.

An exponent below 1 is supported by what the split leaves out. Anderson et al. (2018) recorded the angle a shot was taken from, not the kick that was used to take it. Dwyer and Young (2024) analysed 43,254 shots and found that angle and shot type together determine the outcome, so the angle alone does not. The drop punt is the conventional technique from directly in front, but from a wide angle, a player may instead select a different kick type, such as a snap or a banana kick, suited to their position. Setting $p = 1$ would assume the opposite: that the kick stays fixed and only the target narrows. The exponent is the model's allowance for that choice.

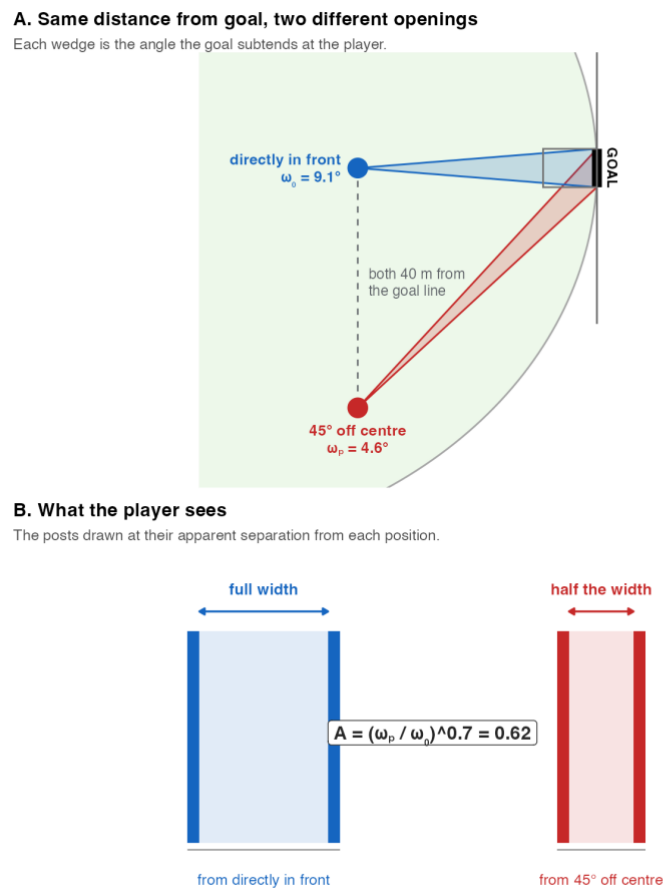


Figure 4. The angle of opportunity. (A) Two players 30 m from the goal line, one directly in front and one 45 degrees off-centre. Each shaded wedge is the angle the goal subtends at that player. (B) The same two openings drawn as the player sees them.

3.1.3 Reachability (R)

Equation 2 was fitted to shots that were taken. It therefore describes how often such a shot is converted, not whether a shot is available at all. Extrapolated to 90 m it returns 18.7%, which no player achieves because a kick of that length does not reach the goal.

The point at which a shot ceases to be an option is not sharply defined. A kick of 55 m is routine for many players, 65 m is exceptional, and beyond 75 m the ball will not carry regardless of technique. A third term therefore reduces Location smoothly with range:

$$R(d) = \frac{1}{1 + e^{(d-d_{50})/s}} \quad \text{Equation 4.}$$

The constant d_{50} is the distance at which reachability falls to one half, and s controls how abruptly it falls. Neither value can be drawn from conversion data because the published data stops before these distances are reached.

That absence is itself informative. Anderson et al. (2018) leave their final band open above 50 m with nothing beyond it. Browne et al. (2022), analysing 9,725 shots from the 2017 season, restricted their analysis to 0 to 60 m and grouped longer attempts together because shots in those ranges were infrequent. The two largest published studies of Australian Football goal kicking therefore end between 50 m and 60 m, not by choice of scope, but because too few shots are taken beyond that range to support analysis.

Reachability is set accordingly. With $d_{50} = 66$ m, the midpoint of the decline sits just beyond the range where shots are recorded frequently enough to be measured. Setting $s = 5$ spreads that decline across roughly 20 m, so reachability retains 90% of the location value at 55 m and 10%

of it at 77 m. Combined with Equation 2, the model gives a goal probability of 33.5% at 55 m, 14.7% at 66 m and 2.4% at 77 m for a shot directly in front.

Location is the product of the three terms:

$$LO = \sigma(a + b \ln d) \times A \times R(d)$$

Equation 5.

Pressure (PR)

A defender standing close to the ball carrier reduces the value of the position without changing where the ball actually is. The Pressure term captures that reduction. Pressure is not a probability and is not added to the location term; it is a discount applied to it, so a carrier under heavy pressure retains only a fraction of the danger the same position would carry if unopposed.

Pressure is evaluated for every defender on the ground and collapsed into one value for the frame.

$$PR = I_{gen} \left(1 - e^{-k_3 \sum_{i=1}^{18} \max\left(0, 1 - \frac{d_i}{r_i}\right)} \right)$$

Equation 6.

Equation 6 is built from one idea, repeated eighteen times and then combined. The leading term, I_{gen} , acts as a switch: pressure applies in general play and is set to zero on a set shot, where a protected possession prevents any defender from closing on the kicker. Inside the sum, every defender on the ground is given a reach, r_i , and scored on how far inside that reach has already got. A defender standing on top of the carrier scores 1. A defender at the very edge of his reach scores 0, and the max function holds anyone further away at 0, so no separate cutoff is needed. Reach isn't the same for every defender: it depends on where he's positioned relative to the carrier and whether he's closing the gap. The notation table sets out its two components.

The head-on radius sets the scale for the other two. A defender only threatens the carrier if he can reach him before the ball is disposed of, so the radius is the distance he can cover in that window. Gastin et al. (2013) tracked professional players at the moment of a tackle and recorded velocities averaging 4.06 m/s across low, medium, and high intensity contacts. Over a disposal window of 1.5 s, that gives 6.0 m. A defender's resting reach depends on his position relative to the line from the carrier to goal: 6.0 m head-on ($\alpha \leq 45^\circ$), 4.5 m lateral ($45^\circ < \alpha \leq 135^\circ$) and 4.0 m behind ($\alpha > 135^\circ$). A defender in front is worth the most because he can both tackle and smother, threatening the kick as well as the possession. A defender beside the carrier can still smother but does not stand between him and the goal, so his radius sits between the other two. Link et al. (2016) gave soccer's smallest radius to defenders behind the attacker, who can neither block the shot nor make legal contact without difficulty. Australian Football imposes fewer restrictions. A tackle from behind is legal provided the player is not pushed in the back (AFL, 2026, p. 14), so a rear defender can end the possession outright. He still cannot smother, which requires body position in front of or lateral to the ball. The rear radius was therefore raised from 3.0 m to 4.0 m to reflect the legal tackle, and held below the lateral radius to reflect the smother he cannot make.

A defender's influence also depends on how fast he is moving, not just where he stands. A fixed radius captures position and ignores speed, so a stationary defender and one bearing down from the same spot would score identically in every sector. Reach is therefore scaled by Δv_i , the defender's speed minus the carrier's. This is a difference, not a ground speed. A defender running at 7 m/s beside a carrier at the same pace is holding station and earns no extra radius; while one

moving at 4 m/s pursuing a carrier at 1 m/s is closing the gap and does so. Only positive values scale the radius, so a defender being outrun gains nothing, and the term reduces to the resting values when no one is gaining. The reference speed $v_0 = 3$ m/s is set against the gains actually observed: 53% of defenders are gaining ground on the carrier, by 1.9 m/s on average and 4.3 m/s at the 95th percentile. With $\lambda = 1$, a defender gaining at 3 m/s doubles his radius, and the fastest closers in the data reach roughly 2.4 times their resting radius. Figure 6 illustrates this correction behind the carrier, where it matters most, though it applies equally head-on and laterally.

The eighteen contributions are added because two opponents converging on a carrier apply more pressure than one alone. They are then passed through $1 - e^{(-k_3x)}$, which grows quickly at first and flattens. This matches how additional defenders behave in practice. The first opponent to arrive removes the carrier's time; while a third arriving to an already lost contest adds little. Saturation also keeps PR inside 0 to 1, so the discount can never invert the location term.

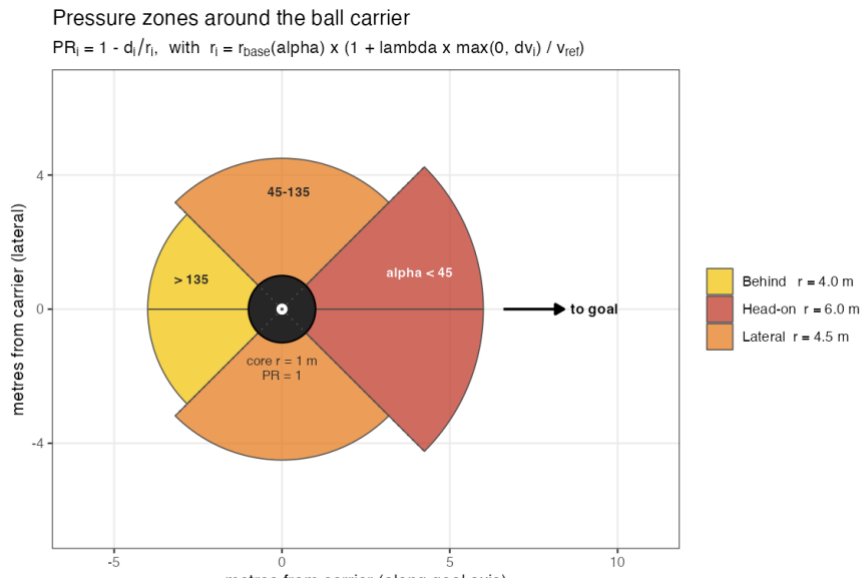
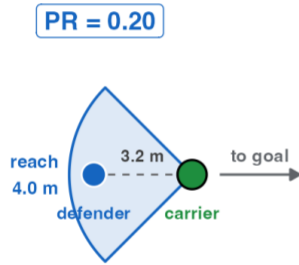


Figure 5. Defender reach around the ball carrier, shown at rest. Reach depends on the sector the defender occupies relative to the line from the carrier to goal: 6.0 m head-on ($\alpha \leq 45^\circ$), 4.5 m lateral ($45^\circ < \alpha \leq 135^\circ$) and 4.0 m behind ($\alpha > 135^\circ$).

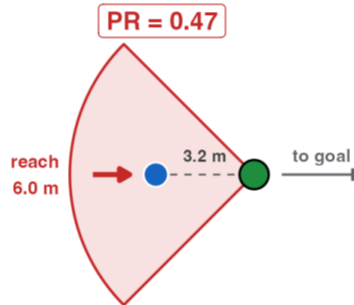
The chase-down: same gap, three speed differences

A defender 3.2 m behind the carrier. The gap never changes. Δv_i is the defender's speed minus the carrier's, not his speed over the ground.

Same speed as the carrier



1.5 m/s faster



3.0 m/s faster

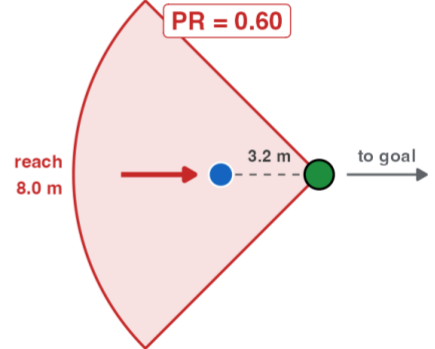


Figure 6. The effect of closing speed on a single defender's contribution. The defender is 3.2 m behind the carrier in all three panels.

Δv_i is his speed minus the carrier's, so it measures whether he is gaining ground rather than how fast he is running. Zero and negative values leave the radius unchanged at its resting 4.0 m.

Structure (ST) and Shot Intent (SI)

A carrier 25 m from goal is close enough for the position to look dangerous. If he is on the boundary line, under pressure, facing eight opponents with only three teammates inside the forward 50, it's not so dangerous for the opposition. Structure measures how space is occupied. Location cannot capture this, because it describes only where the ball is, nor can Pressure since it only counts the defenders near the carrier.

Structure does not apply equally to every frame. A player lining up 20 m out, directly in front, is not looking for a teammate, and the number of opponents inside the forward 50 has no bearing

on what he does next. A player on the boundary line at the same distance, however, will likely be looking to pass, and that decision shapes the play. Shot Intent separates these two cases. It functions not as an independent penalty but as a gate: the closer a frame is to a shot, the less the structural discount applies, which is why Structure enters the model as $(1 - SI) \cdot ST$ rather than on its own.

The two quantities are evaluated separately. Structure is computed from the players inside the forward 50 and Shot Intent from the carrier's own position.

$$ST = \left(0.5 + \frac{\arctan(k_5(n_D - n_A))}{\pi} \right) (1 - e^{-n_D/n_{ref}}) \quad \text{Equation 7.}$$

$$SI = R(d)[1 - I_{gen}(1 - \cos \theta)] \quad \text{Equation 8.}$$

The first line has two brackets, each answering a different question. The first asks who holds the numbers: n_D defenders inside the forward 50 against n_A attackers, excluding the carrier, who cannot be his own kicking target. The forward 50 is used because it is a fixed, marked boundary rather than a chosen radius. Structure measures occupancy of that region, not obstruction of a direct line to goal. Link et al. (2016) modelled the latter as the probability of a shot being blocked, which has no counterpart here, because an unbounded target cannot be obstructed in the same way. The arctangent maps their numerical difference onto a bounded scale centred on an even contest, so equal numbers return exactly 0.50, a defence three players up returns 0.67, and three players down returns 0.33. The constant $k_5 = 0.2$ sets how sharply the value responds, and the bound prevents any mismatch from driving Structure to an extreme. The second bracket asks how many defenders are present at all, rising from 0 with none inside the arc to 0.63 at three and

0.98 at twelve, with $n_{ref} = 3$ setting the rate. Structure is the product of the two brackets, so it is high only when the opposition both outnumber the attack and is present in force.

The second line asks how much the frame resembles a shot rather than a disposal. $R(d)$ is reachability, already defined in Equation 4: a ball that cannot carry to the goal is not a shot from anywhere on the ground. The cosine falls from 1 directly in front to 0 at 90 degrees off centre, so a carrier in the pocket is treated as less likely to be shooting, regardless of his distance from goal. A player 30 m out and straight in front returns a value near 1.00, the same player at 45 degrees returns 0.71, and the 25 m frame described above returns 0.17. The indicator removes the cosine on a set shot, because a player awarded a mark or a free kick has already been given the kick, and the angle no longer decides whether he takes it.

A numbers advantage alone cannot describe a forward line. M is zero on 21 of the 102 frames in the tested dataset, and those frames hold anywhere between 0 and 12 defenders inside the arc. An empty forward 50 and a twelve-on-twelve contest are both even, yet only one of them obstructs a kick. The second bracket in the ST equation therefore scales the advantage by how many defenders are present, rising from 0 with none to 0.63 at three and 0.98 at twelve, with $n_{ref} = 3$ setting the rate. Across those 21 frames, Structure now spans 0 to 0.49 rather than returning 0.5 for every one of them.

Limitations and Extensions

The model's limitations follow from what a single frame of tracking data can hold. Speed is recorded as a magnitude rather than a vector, so Δv compares two numbers rather than measuring whether a defender is closing, and a defender running away from the carrier stretches his reach just as much as one bearing down on him. The frame also carries no history, so there is

no possession clock, no contest state, and no record of whether a mark was taken cleanly or won in a pack; a player with time and space is scored identically to one who has just emerged from a contest. Nor does it carry match conditions: in the wet, for instance, the ball is less likely to be moved cleanly out of the forward 50, and territory is worth more than the model allows. These are limitations of the input data rather than of the formula itself, and each would be resolved by richer data rather than a different equation.

Player direction would convert Δv into a true closing speed and resolve the first limitation directly. A larger change would be to replace goal probability with an expected score, since a behind is worth a point and the model currently counts it as nothing; this requires shot outcomes by location. Whether a behind is worth having then depends on match context, because a scenario where scores are level in the last minute carries different value than one where a team is six goals ahead, so score and time remaining would be incorporated as well t . Player identity and skill would separate players the model currently treats as interchangeable. A frame carries position and speed alone, so a small forward running at goal from 30 m and a ruckman arriving at the same point return the same value, despite representing different levels of danger. A per-player term would require both player identity and a measure of goal-kicking skill, neither of which the current data supplies.

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